# ANALYZING E -LEARNING PLATFORM PURCHASES USING MYSQL

**Objective**

Based on my **LearnHub** dataset and analysis, here are the **key business insights**.

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**1. Revenue Performance by Category**

- **Programming** generated the highest revenue due to the high price of programming courses and larger purchase quantities.
- **Analytics** showed strong learner interest with multiple course enrollments across different learners.
- **Database** courses maintained steady sales, driven mainly by the popularity of SQL Basics.

**Insight:** Programming courses contribute the most to revenue, making them the platform's primary revenue driver.

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**2. Learner Engagement**

- Several learners purchased more than one course, indicating repeat learning behavior.
- Learners from different countries participated, showing international reach.
- Some learners have not made any purchases, as seen in the LEFT JOIN results.

**Insight:** While active learners show strong engagement, there is an opportunity to convert inactive registered learners into paying customers.

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**3. Most Purchased Courses**

From the Top 3 Courses analysis:

1. Python Basics (Quantity: 4)
2. SQL Basics (Quantity: 3)
3. SQL Basics (Quantity: 3)

**Insight:** Technical skill-based courses, especially Python and SQL, are the most demanded learning products.

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**4. Cross-Category Learning Behavior**

The "Learners Purchasing from Multiple Categories" query identifies learners who bought courses from more than one category.

**Insight:** These learners are highly engaged and represent ideal candidates for premium subscriptions, bundles, or personalized recommendations.

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**5. Course Popularity**

- SQL Basics and Python Basics received the highest purchase quantities.
- Analytics courses such as Power BI and Tableau Fundamentals also attracted multiple learners.

**Insight:** Data and analytics skills are highly valued by learners.

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**6. Untapped Course Opportunities**

The "Not Purchased Courses" query reveals courses that have never been purchased.

Likely:

- Java Programming
- Web Development

**Insight:** These courses may require:

- Better marketing
- Discount campaigns
- Updated content
- Improved course descriptions

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**7. Purchasing Trends**

- Purchases occurred consistently between January and May 2026.
- No major drop in enrollments was observed during this period.

**Insight:** The platform maintained steady learner activity throughout the analyzed timeline.

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**8. Customer Spending Analysis**

The Total Spending analysis shows:

- Learners purchasing larger quantities significantly increase revenue.
- A small number of learners contribute a large share of total sales.

**Insight:** Identifying high-value learners can help create loyalty and retention programs.

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**9. Geographic Distribution**

Learners come from:

- USA
- India
- Canada
- Spain
- UAE
- UK
- Australia
- Egypt

**Insight:** LearnHub has a diverse global audience, creating opportunities for region-specific promotions and localized content.

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**10. Business Recommendations**

✅ Promote Programming courses since they generate the highest revenue.

✅ Create course bundles combining SQL, Python, and Power BI.

✅ Offer discounts for inactive learners who have registered but not purchased.

✅ Improve visibility and marketing of courses with zero purchases.

✅ Introduce loyalty rewards for repeat learners and multi-category purchasers.

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**Conclusion**

The **LearnHub** analysis revealed that Programming and Analytics courses contribute significantly to platform revenue and learner engagement. The study also identified high-value learners, top-performing courses, and underutilized offerings, enabling data-driven decisions to improve course marketing, learner retention, and overall business growth.